ADBC Scanner

Database drivers. Arrow batches. Ordinary SQL.

Install & load

```
SET allow_extension_repositories = 'allowed';
CREATE EXTENSION REPOSITORY IF NOT EXISTS query_farm WITH PREFIX 'https://duckdb-extensions.query.farm';
INSTALL adbc_scanner FROM query_farm;  LOAD adbc_scanner FROM query_farm;
```

01 At a glance

Connect DuckDB to a database through an Arrow Database Connectivity driver. Use attached catalogs for joins and discovery, send backend SQL explicitly, or bulk-ingest a DuckDB query through the driver's Arrow interface.

ADBC driver → Arrow batches → DuckDB SQL

02 ATTACH API essentials

`ATTACH … (TYPE adbc)`

Open a catalog; pass its alias to the functions below.

`adbc_scan(db, sql)`

Execute backend SQL; supply params and columns when needed.

`adbc_scan_table(db, table)`

Read a table with schema discovery and supported pushdown.

`CALL adbc_execute(db, sql)`

Run backend DDL or DML; return affected-row count.

`adbc_insert(db, table, query)`

Ingest Arrow batches; modes include create, append and replace.

`adbc_schema(db, table)`

Describe a named table, rather than an arbitrary query.

`CALL adbc_clear_cache()`

Refresh attached catalog metadata after remote DDL changes.

`SET adbc_allowed_drivers = 'sqlite'`

Allow only listed drivers; profiles and search_paths are then refused.

03 Keep in mind

Sandbox the drivers

ATTACH loads a native driver library into the process and is refused while enable_external_access is false. Set adbc_allowed_drivers, then lock_configuration, before running untrusted SQL. Install the ADBC driver manifest or shared library separately.

04 Query through a driver

MANUAL SETUP

```
ATTACH ':memory:' AS db (
  TYPE adbc, driver 'sqlite'
);
SELECT * FROM adbc_scan(
  'db', 'SELECT ? AS id',
  params := row(42),
  columns := {'id': 'BIGINT'}
);
```

Requires an installed SQLite ADBC driver. Returns id = 42.

05 Bulk load a DuckDB result

MANUAL SETUP

```
SELECT * FROM adbc_insert(
  'db', 'people',
  (SELECT 1::BIGINT AS id, 'Ada' AS name),
  mode := 'create'
);
SELECT * FROM adbc_scan_table('db', 'people');
SELECT * FROM adbc_schema('db', 'people');
DETACH db;
```

Continues the first example; ingestion modes depend on the driver.

06 Restrict which drivers load

```
SET adbc_allowed_drivers = 'sqlite';
SELECT current_setting(
  'adbc_allowed_drivers') AS allowed;
-- sqlite
```

An ATTACH naming any other driver, profile or search path now fails.

07 In practice

Schemas and transactions

SQLite needs explicit columns for arbitrary scans. BEGIN / COMMIT controls the attachment's remote transaction, not a distributed one. Use a file for SQLite catalog scans. Cancelling a query cancels the remote statement; passwords show as *** in duckdb_databases() and errors.

Full docs & examples ↗ Source ↗